THE SI SENTINEL

Weekday briefing on builders shaping super intelligence

TUESDAY, OCTOBER 13, 2026

THE MONDAY PICTURE

Meta reaches the frontier, and everyone wants their own chips

Mark Zuckerberg and Alexandr Wang released Muse Wonder on Monday, the first frontier model from Meta Superintelligence Labs. Within hours Tobi Lütke said he could hardly find a task it couldn't do, Aaron Levie said a new player at the top would push token prices down, and David Heinemeier Hansson said he had already shipped code with it. The launch was messy too: a quota bug drained paid plans, and Wang admitted the model is "not as good as opus 5.5 on coding for sure." Elon Musk said the numbers being passed around "do not match at all what we see with Grok @Bot!"

On silicon, Satya Nadella made Microsoft's "biggest bet yet" on its own chips with Maia 300, and Lisa Su showed AMD's Helion rack and Verano processor for agent workloads. Broadcom's Hock Tan called many AI investors casino gamblers, and Musk answered, "No crying in the casino." Uncle Bob Martin put his agent productivity gain at "about a factor of five," Demis Hassabis cheered a Lancet paper on a diagnostic AI used with real patients, and Palmer Luckey unveiled Anduril's Steelfin interceptor. Musk posted 31 times; 23 of the 45 builders posted.

I · THE NEW FRONTIER LAB

Muse Wonder arrives, bugs and all

Meta's superintelligence lab shipped a model that scores near the top of independent rankings, then spent the evening fixing its launch.

56
Muse Wonder Full's Artificial Analysis Intelligence Index score at max effort, up from 48 for Muse Spark 1.3 six weeks ago

$4 / $20
Muse Wonder API price per million input and output tokens

1M
Context window in tokens for both new models

Mark Zuckerberg and Alexandr Wang @finkd · CEO, Meta · @alexandr_wang · chief AI officer, Meta

At 12:55pm EDT Zuckerberg posted: "Introducing Muse Wonder, our new family of frontier models with state of the art performance for personal and professional agentic work." It is the lab's second model family, after the smaller Muse Spark:

> *"It's been just over a year since we started Meta Superintelligence Labs. Muse Spark was our first family of models, and now Muse Wonder is a much larger and more advanced class of models that brings us to the frontier."*
>
> MARK ZUCKERBERG · @FINKD · POST

There are two models. Muse Wonder is sold through the Meta Model API and the Muse Code coding tool at $4 per million input tokens and $20 per million output tokens. The larger Muse Wonder Full is available only to Muse Code subscribers on the Power and new Maximum plans, Meta's developer blog says. Both were trained at Meta's gigawatt-scale Prometheus data center, according to the research blog, and will power the Muse assistant, starting with subscribers. TechCrunch called it Meta's first attempt at a frontier model since Wang restructured its AI work in August 2025.

Wang pitched the model on personal agents, with "ACTIVE MEMORY: best memory across days/weeks/months" and "ACTIVE FOCUS: gets your tasks done." He said the team built an internal Personal Superintelligence Index to measure "what actually makes a model useful for personal agents, vs just focusing on external benchmarks," and that the model "certainly has big model smell (which we quantified!)."

The independent benchmarking firm Artificial Analysis scored Muse Wonder Full at 56 and Muse Wonder at 52 on its Intelligence Index. The biggest gains were in agent tasks: Terminal-Bench 4.0 rose from 33 percent to 57 percent. On its coding-agent index, Muse Wonder Full in Muse Code scored 68, level with Claude Code running Sonnet 5.5. The firm also found the API model relatively costly to run, at $4.52 per index task.

QUICK HIT

Cheeky Pint. Zuckerberg and Wang took Muse Wonder to John Collison's pub interview show. Wang said that at one point he "get my annual performance review live on camera -- a truly tender moment."

The builders weigh in @tobi · @rauchg · @levie · @dps · @dhh

Lütke, CEO of Shopify, had early access. His verdict drew nearly 5,000 likes:

> *"This model is very very good. I was struggling to find anything to throw at it that it couldn't do, even when other top models failed. Meta is back."*
>
> TOBI LÜTKE · @TOBI · POST

Guillermo Rauch, CEO of Vercel, called it "Incredibly fast progress by Meta" and put it on Vercel's AI Gateway the same afternoon. Levie, CEO of Box, saw a price effect: "Now that there are multiple players pushing the state of the art, this is going to lead to a further acceleration of advances, as well as the lowering cost of tokens for even the highest levels of capability." David Singleton, the former Stripe chief technology officer, said he had been using it in his own Muse for a while and liked how it "works harder and draws on memory more precisely." Heinemeier Hansson reported "I've already shipped two Omarchy PRs built with Wonder!" and said 37signals "will be using a lot of Wonder with our $1.5m @AIatMeta token stash."

Launch-day trouble @alexandr_wang · @elonmusk

Within two hours, subscribers on the $50 plan reported that a few prompts had used up their five-hour limit. Wang posted that Meta was "working to resolve a bug in how muse wonder and muse wonder full are counting against usage limits," and at 6:31pm EDT announced a fix and a full quota reset. Early users posted weak 3D demo games, and Wang answered:

> *"whole team feels dumb now, but it's true, we did not explicitly train our model on 3d demo games. obviously in retrospect we probably should've done a bit of that... but no regrets, please try muse wonder on your actual coding work!"*
>
> ALEXANDR WANG · @ALEXANDR_WANG · POST

Asked whether Zuckerberg had overhyped the benchmarks, he conceded: "it's not as good as opus 5.5 on coding for sure." Musk was skeptical. When Garry Tan, CEO of Y Combinator, shared figures from investor Deedy Das and wrote "My bet would be Muse is going to win," Musk replied, "These numbers do not match at all what we see with Grok @Bot!" To a tester who called the model "benchmaxxed," meaning tuned to benchmarks, he wrote "Hmm." Others were warmer. The developer and YouTuber Theo Browne ranked Muse Wonder in his top tier beside Opus 5.5, and Wang replied, "wow, S tier from theo! not on my bingo card."

II · SILICON

Everyone builds the stack

Microsoft and AMD showed new hardware for the age of agents, while Broadcom's chief warned that the money behind it is getting reckless.

40%	7x	32,768
More tokens per megawatt Microsoft expects from Maia 300 than from leading GPU systems	Rack-scale token throughput of a 96-chip Maia 300 rack versus Maia 200	CPU cores in a CPU-only AMD Helion rack for running agents

Satya Nadella and Mustafa Suleyman @satyanadella · CEO, Microsoft · @mustafasuleyman · CEO, Microsoft AI

Nadella opened Monday with a hardware announcement:

> *"Today we are announcing our biggest bet yet on our own silicon. Starting next calendar year, we will scale our Maia family to multi-gigawatt deployments, making it the primary inference system for our own product and services. And we're introducing Maia 300, which delivers 40% more tokens per watt compared to leading GPU-based systems."*
>
> SATYA NADELLA · @SATYANADELLA · POST

He added that the current Maia 200 is already deployed across the US, Europe and Australia and delivers up to 1.7 times more tokens per watt in Teams and Copilot. In the official announcement, cloud chief Scott Guthrie said a Maia 300 rack packs 96 accelerators into 167 kilowatts with 27.6 terabytes of high-bandwidth memory (HBM), and that production deployments begin in 2027. SiliconANGLE reported that each chip uses four compute chiplets on TSMC's 2-nanometer process with 288 gigabytes of HBM4, the same capacity as Nvidia's Rubin but with about 20 percent less bandwidth. Suleyman, who runs Microsoft's consumer AI group, quoted the post: "40% more tokens per watt!"

The investor Chamath Palihapitiya read the news as chip companies moving up the stack while application companies move down, so that "everyone competes with everyone." Musk, who is building his own chip plant, Terafab, named the hard part: "Scaling power harnessed, along with super fast construction, cooling, networking and batteries. This is hard."

Lisa Su @LisaSu · chair and CEO, AMD

At the Open Compute Project (OCP) Global Summit in San Jose, Su gave the opening keynote. Afterward she wrote: "A year ago we showed Helios and now we're ramping Helios in production. Today we introduced Helion, an open, flexible rack for agentic AI and showed 'Verano' - our newest agentic CPU for the very first time." Helion is a reference design, not a product AMD will sell itself. Its CPU-only version holds up to 32,768 EPYC cores to run agents next to separate inference hardware. An all-in-one version adds Instinct MI350P graphics processors, and a built-in router sends agents' model calls to frontier or open models, AMD said. Partner configurations are due in November, and AMD will contribute the spec to OCP. Verano, the first EPYC chip with a low-power LPDDR memory interface, ships in 2027.



Frontier, the exascale supercomputer built by Hewlett Packard Enterprise with AMD processors, at the Energy Department's Oak Ridge National Laboratory in Tennessee, June 2022.

PHOTO: OLCF AT ORNL · CC BY 2.0 VIA WIKIMEDIA COMMONS

Elon Musk @elonmusk · on memory and the "casino"

Memory-chip stocks fell after China's ChangXin Memory Technologies (CXMT) announced progress on a denser DRAM design. One investor called the selloff senseless. Musk was more cautious:

> *"When China figures out how to manufacture a product, their ability to scale is often greater than the rest of the world combined. We saw this with lithium batteries and solar, for example. SpaceX is a rare counter-example, where its mass to orbit is roughly an order of magnitude greater than the rest of the world, BUT China now does a majority of non-SpaceX mass to orbit. SpaceX is an anomaly in the matrix."*
>
> ELON MUSK · @ELONMUSK · POST

"Maybe RAM will be different," he went on, noting that the LP5 and LP6 standards are ones "China seems to be solving fast," while newer HBM designs that stack memory on top of logic and keep it cool "will be harder to solve." The same day, at a DBS conference in Singapore, Broadcom CEO Hock Tan said "a lot of investors have no clue what they're investing in," that "it's almost like they are in a casino," and that companies burning AI tokens are like children spending $10 at an arcade in 10 minutes. South Korea's Kospi fell about 6 percent on Tuesday, The Economic Times reported, with Samsung and SK Hynix down as much as 9 percent. Musk's first response was "No crying in the casino." To the investor Michael Burry, who posted the arcade line, he wrote: "Really fun way to spend $10 as a kid tbh. I loved it! And you do learn things if you play many video games."

III · AGENTS AT WORK

Every morning is Christmas

Veteran programmers described what working with agents is really like, and Musk spent the evening promoting Grok Bot.

Uncle Bob Martin @unclebobmartin · author, Clean Code

Robert Martin now uses an agent to edit his videos. It "listens to them, sorts them, compares them to the script, let's me know of any irregularities, and then places them in the Camtasia timeline for me." He added, "shooting scenes in the Cybertruck is a breeze. The car drives while I record." That afternoon he found himself talking to Grok Bot on an iPad while it changed files on his Mac to fix an audio sync problem: "It's like talking to a very capable and thoughtful assistant. Wierd."

In a long exchange with a skeptic, he said agents build in 30 minutes what would take him a day or two, then he spends about an hour on quality tools and manual testing:

> *"To be conservative, I'd say it's about a factor of five in productivity. It could be more. And that's when I'm using just one agent."*
>
> UNCLE BOB MARTIN · @UNCLEBOBMARTIN · POST

"In the last month or two the models have gotten MUCH better," he wrote. "It's both uncanny and unsettling." When the software engineer Grady Booch called one of his replies "profoundly useless," Martin explained the catch: "Modern models have significant software engineering skills. But modern models don't apply them proactively. Ask a model to do X and it'll do X the way a novice would." Ask the model to review that work, he added, and it will list the engineering problems it ignored.

David Heinemeier Hansson @dhh · creator of Ruby on Rails, co-owner of 37signals

Heinemeier Hansson, who is in Japan for an Omarchy meetup in Tokyo, posted a series of notes on agent-era maintenance. "Every morning is Christmas now: I wonder what gifts Agent Santa brought to me in the night!" He now prefers bug reports and patches from agents for Omarchy, his Linux setup:

> *"Rare is the human who has the patience to match an agent in providing all the pertinent details in a bug report. Also, PRs are mostly just proofs of concept now anyway."*
> DAVID HEINEMEIER HANSSON · @DHH · POST

The flip side, he added, is that "the backlog is going to be agentically managed too." He said agents have cut the cost of "obsessive performance tuning" to "near zero," and announced Signed, a PDF-signing app that replaces Xournal++ in Omarchy and was "nearly a perfect one-shot prompt." He also praised Linus Torvalds, who wrote in a kernel tag that he "did what I preach" and used AI tools to untangle obscure code; a test harness then found three bugs dating to 2013. Heinemeier Hansson revisited a clip of Dario Amodei, CEO of Anthropic, predicting that models could do most software engineering end to end within 6 to 12 months: "Dario turned out to be right. Maybe even a little cautious." And when Tim Sweeney said Epic Games is moving developers around "for maximum AI acceleration," he answered, "This is the way!"

QUICK HIT

Paul Graham @paulg, co-founder of Y Combinator: "When I suggested he might want a more outgoing cofounder to do sales, he said he didn't, because most of his sales would be to agents. I had to admit he could be right."

Aaron Levie and Guillermo Rauch @levie · CEO, Box · @rauchg · CEO, Vercel

Levie predicted that swarms of agents will become "one of the biggest token uses in large enterprises," with cybersecurity an early winner, because testing huge numbers of attack combinations "without getting tired or bored" is exactly what agents do well. He also endorsed new research, flagged by computer scientist Pedro Domingos, in which the model designs its own harness: "In the limit, AI will obviously be better at determining its own harness once it sees the task at hand, than us being able to do this in advance." Rauch argued for less: "Coding CLIs keep getting more complicated. Daemons, 3 layers of TUIs, client-server architectures." Vercel's fx tool, he said, is 6 megabytes, "starts in milliseconds" and works with any model. One fan ported it to Windows 98, talking to a language model over a 56k modem. Rauch: "This is amazing."

Elon Musk @elonmusk · on Grok Bot

Most of Musk's posts were short endorsements of users praising Grok Bot. One user wrote that it had replaced all of their search-style chatbot queries and now runs their Cursor cloud agents, and Musk wrote "Grok @Bot rocks." Quoting a consultant who said he had brought 71 corporate clients onto it, he wrote "Grok @Bot is like hiring a super smart employee." He also shared Memory Transfer, a Grok Bot that imports what ChatGPT or Claude learned about a user and asks before saving each item. When Emad Mostaque, founder of Stability AI, said he kept hitting his usage limit, Musk asked: "What are the top things we need to fix or features we need to add to Grok @Bot?"

QUICK HIT

Kanjun Qiu @kanjun, CEO of Imbue, on early reviews of Imbue Studio, its tool for building personal apps: "my Studio apps are starting to replace email/chat apps."

IV · SCIENCE AND MEDICINE

From the lab to the clinic

A diagnostic AI passed a real-patient test, mathematicians argued over machine-made proofs, and open researchers asked for help.

Demis Hassabis @demishassabis · CEO, Google DeepMind

Hassabis quoted Sundar Pichai's post on Google's first paper in The Lancet's main journal: "Amazing to see research move from the lab to real-world clinical utility - congrats to the team on this milestone & @TheLancet paper!" The study tested AMIE, a research chatbot that patients talk to before an appointment, with 98 urgent-care patients at Beth Israel Deaconess Medical Center in Boston. Pichai wrote that clinicians said its summaries helped them prepare in 75 percent of cases and changed their approach to care in more than half, and that AMIE's list of possible diagnoses matched doctors' final diagnoses 90 percent of the time.

Noam Brown @polynoamial · researcher, OpenAI

The mathematician David Bessis posted "10 remarks on AI and math that can be true all at once." He called recent progress "stunning and exciting" and said "mathematics is being transformed and there is no way back." His fourth remark said, in blunter language, that OpenAI does not care about the future of mathematics. Brown answered that point:

> *"Many of the OpenAI researchers behind these results are math PhDs or former math professors. If anything, they care too much about the future of mathematics. AI is disruptive but we also think it will be an invaluable tool for the future of mathematics."*
>
> NOAM BROWN · @POLYNOAMIAL · POST

Open science @ClementDelangue · @JeffDean · @VitalikButerin

Clément Delangue, CEO of Hugging Face, followed up on the weekend's open math environments with a plea: "If you're a researcher, please consider sharing your RL environments openly! Something you built for one experiment could help the entire field move forward. I know we're in dark days for sharing from frontier labs in the US." Jeff Dean, Google's chief scientist, answered a fundraising appeal from Nando de Freitas for Deep Learning Indaba in Africa and Khipu in Latin America, which are struggling to raise money this year. Dean said his family foundation has supported both and that Discovery Loop "is thrilled to support!" He asked people at AI companies to push their employers to sponsor too. The next Khipu meeting is April 5 to 9, 2027, in São Paulo. Vitalik Buterin, co-founder of Ethereum, recommended PopVax, a vaccine developer with an "open-source open-science and public-spirited approach," which has entered clinical trials.

François Chollet and Elon Musk @fchollet · @elonmusk · on self-improvement

Chollet, who argued last week that self-improving systems still deliver linear progress, offered a reminder from the farm:

> *"If you explained agriculture to folks in SF, you'd blow so many minds. It's... self-replicating, recursively self-improving nanotech for the physical world. And it's all open-source!"*
>
> FRANÇOIS CHOLLET · @FCHOLLET · POST

Early Tuesday, Rob Wiblin of the 80,000 Hours podcast wrote that Jack Clark, co-founder of Anthropic, now thinks full recursive self-improvement (RSI) is more likely to become possible in 2027 than in 2028. In May Clark had put about 30 percent odds on the end of 2027 and 60 percent on the end of 2028, The Decoder reported. Musk's reply: "Obv."

V · MACHINES IN THE WORLD

Cheaper interceptors, longer robotaxi days

Anduril pitched a missile it says can be mass-produced, and Tesla stretched its driverless service and opened chargers to hurricane victims.

Palmer Luckey @PalmerLuckey · founder, Anduril Industries

As the Association of the US Army (AUSA) annual meeting got under way in Washington, Luckey wrote: "We just unveiled Steelfin at AUSA. It is a modular, low-cost mass-production missile defense interceptor that brings two-stage performance to dozens of existing platforms." Anduril says Steelfin is a family, not a single missile. Interchangeable boosters and swappable seekers let one production line build versions for threats from cruise missiles to ballistic and hypersonic weapons, and the missiles fit existing Patriot, Mk 41 and Typhon launchers, the company said.



A US Army Patriot missile fires during the Balikatan exercise in the Philippines, April 25, 2023. Steelfin is designed to fit existing launchers such as Patriot's.

PHOTO: US MARINE CORPS/CPL. TYLER ANDREWS · PUBLIC DOMAIN VIA WIKIMEDIA COMMONS

Executives told Breaking Defense they aim to sell it for about a third of the price of the Patriot's PAC-3 MSE interceptor. Flight tests begin next year, with full-rate production expected at the end of 2028 or early 2029, at "multiple thousands" a year at the low end. Steelfin has not yet flown. When a critic at the show said Ukraine's 3rd Army Corps had not used Anduril products in a recent offensive, Luckey replied that "Anduril has had people and hardware in Ukraine since the second week of the war" and that he had personally trained people in Kyiv and Lviv.

Elon Musk @elonmusk · Tesla Robotaxi and Supercharging

Musk shared Tesla's list of recent "Robotaxi improvements". Service now runs from 6am to 11pm, wait times are shorter thanks to more cars and better dispatch software, the trunk opens automatically at the end of a ride, and Grok Bot is now available inside the Cybercab. The 11pm close in Austin began this month, TeslaNorth reported. When Florida Governor Ron DeSantis thanked Tesla for free Supercharging in areas hit by the hurricane, Musk wrote "Of course." The free charging covers northwest Florida after Hurricane Isaias, TeslaNorth said.

QUICK HIT

Michael Dell @MichaelDell explained how his and Susan Dell's $6.25 billion gift is reaching 25 million children's accounts: first, eligible children born 2016 to 2024 in ZIP codes with median family income of $150,000 or less, then military children overseas, then auto-enrolled children from the lowest-income ZIP codes up.

Takeaways

1. Meta is now a frontier lab. Independent scores put Muse Wonder Full near the top, and a fourth serious competitor should push token prices down.
2. The launch was rough: quota bugs, weak demos and an admission that it trails Opus 5.5 on coding. The real test is daily use over the coming weeks.
3. The biggest AI buyers are designing their own silicon. Microsoft wants Maia to carry most of its inference from 2027.
4. Agents are changing what chips are for. AMD is selling racks of CPUs just to run agents, separate from the inference hardware.
5. Hock Tan's warning hit markets hard. The questions now are whether enterprise AI spending pays off and whether money runs out first.
6. Experienced programmers now describe agents as everyday colleagues, with real but uneven skills, and a factor-of-five speedup.

On the beat

POSTED · 23

Elon Musk, Mark Zuckerberg, Alexandr Wang, Tobi Lütke, Guillermo Rauch, Aaron Levie, David Singleton, David Heinemeier Hansson, Satya Nadella, Mustafa Suleyman, Lisa Su, Uncle Bob Martin, Paul Graham, Kanjun Qiu, Demis Hassabis, Noam Brown, Clément Delangue, Jeff Dean, Vitalik Buterin, François Chollet, Palmer Luckey, Michael Dell, Yann LeCun.

SILENT · 22

Marc Andreessen, Jensen Huang, Ashok Elluswamy, Andrej Karpathy, Sam Altman, Dario Amodei, John Carmack, Patrick Collison, Fei-Fei Li, Greg Brockman, Andrew Ng, Michael Truell, Tom Brown, Jim Fan, Mira Murati, Ilya Sutskever, Sundar Pichai, Joe Gebbia, Nat Friedman, Arthur Mensch, Guillaume Lample, Mark Chen.

Also on the wire

Mark Zuckerberg @finkd, on his own agent: "Muse ordered me [a watermelon] how thoughtful!" Wang has been tagging his Muse Wonder posts with the fruit.

Guillermo Rauch @rauchg on Vercel's scriptc compiler for TypeScript, which can refuse to compile functions that may throw: "We can do better than Rust."

Clément Delangue @ClementDelangue laughed along with three minutes of uncut testing of the small Microduck robot on carpet, faceplant included.

Elon Musk @elonmusk agreed with Nikita Bier, head of product at X, that a $2 billion video game arriving the same month AI "became capable of creating comparable experiences" marks the end of the big-studio era: "Yeah."

Michael Dell @MichaelDell shared a note about James, 8, who "loves math, watches @CNBC... with his mom, and wants to be a CEO," and said he will be on CNBC Tuesday.

Elon Musk @elonmusk, quoting his mother Maye Musk's memory that he always said "Safety First" in Tesla's early days: "Your safety has always been priority 1 at Tesla and SpaceX."

Elon Musk answered "Yes" to a post calling Terafab "a chip factory built for a future where we harness the power of our own star."

Palmer Luckey @PalmerLuckey to Lisa Su on her National Medal of Science: "Congratulations, you deserve this and more!"

Uncle Bob Martin @unclebobmartin on design rules for agents: "Agents like things to be well organized, just like we do. However, I think the thresholds are different."

Alexandr Wang @alexandr_wang, on a user who found Muse Wonder talks more naturally than rivals: "this was a big focus area for muse wonder; our goal was to build a model for everyone!"

Tobi Lütke @tobi cheered Flexport's newest automated fulfillment center outside Chicago, which its CEO announced with "Warehouse robots are here!"

Vitalik Buterin @VitalikButerin on the open-source vaccine group RaDVaC, which is working on oral vaccines made with yeast: "Radvac is excellent! Balvi supported them too early on."

Clément Delangue asked a startup that built similar math environments to add them to Hugging Face.

David Heinemeier Hansson @dhh: Omarchy now offers an optional traditional desktop with a dock, removing "one more excuse not to switch." Its Tokyo meetup on Tuesday is "at-capacity."

The SI Sentinel is compiled from public posts on X and published reporting. Quotes are reproduced verbatim unless marked as translated or paraphrased.